

Lab 1: Diodes

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This lab investigates the behaviors of diodes. They are small semiconductors that only allow current to flow in one direction. A standard diode consists of three parts. A P-type region (positive charge), and N-type region (negative charge), and a junction region (sometimes called the depletion region). A diode can operate in forward bias or reverse bias. Forward bias is when the p-side of the diode is connected to the positive side of a source and the n-side to the negative. The “pressure” of the positive voltage alters the junction region, a barrier, and allows current to pass. When in reverse bias, the p-side is connected to the negative source and the n-side to the positive. The barrier, which is the junction region, increases and therefore stops current from flowing. Only a small amount can pass through until the diode reaches what is called “breakdown.” The threshold in which the diode will not let any current leak through. Diodes are usually made of silicon and used in every electronic device. An example of the general structure is below:

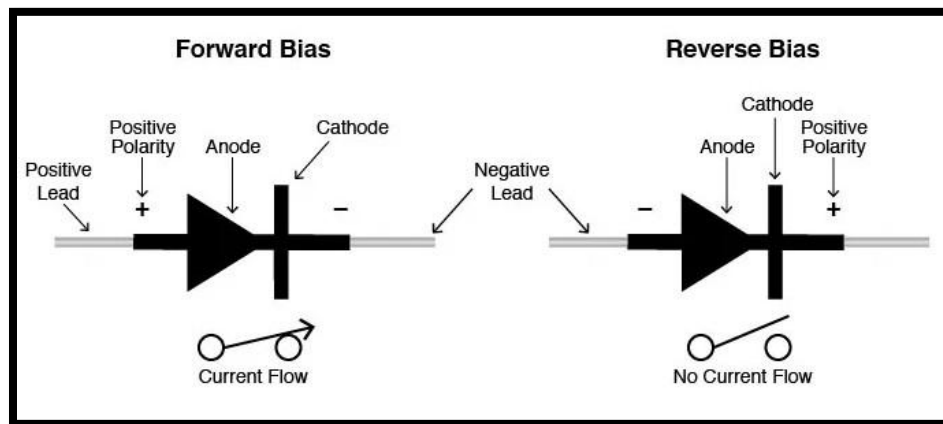


Figure 1: Diode Structure

Fundamentals:

Finding $\frac{k_b T}{q}$ in units of volts at a temperature of $T = 300K$:

$$k_b = 1.38 \times 10^{-23} \frac{J}{K}$$

$$T = 300K$$

$$q = 1.602 \times 10^{-19} C$$

$$V_{th} = \frac{(1.38 \times 10^{-23})(300)}{1.602 \times 10^{-19}} \approx 25.85 \text{ mV}$$

Plotting the ideal diode equation I-V curve for $T = 100K$ to $T = 400K$ was next. I used MATLAB and created subplots so it is legible. Else the 100K curve will dominate the graph.

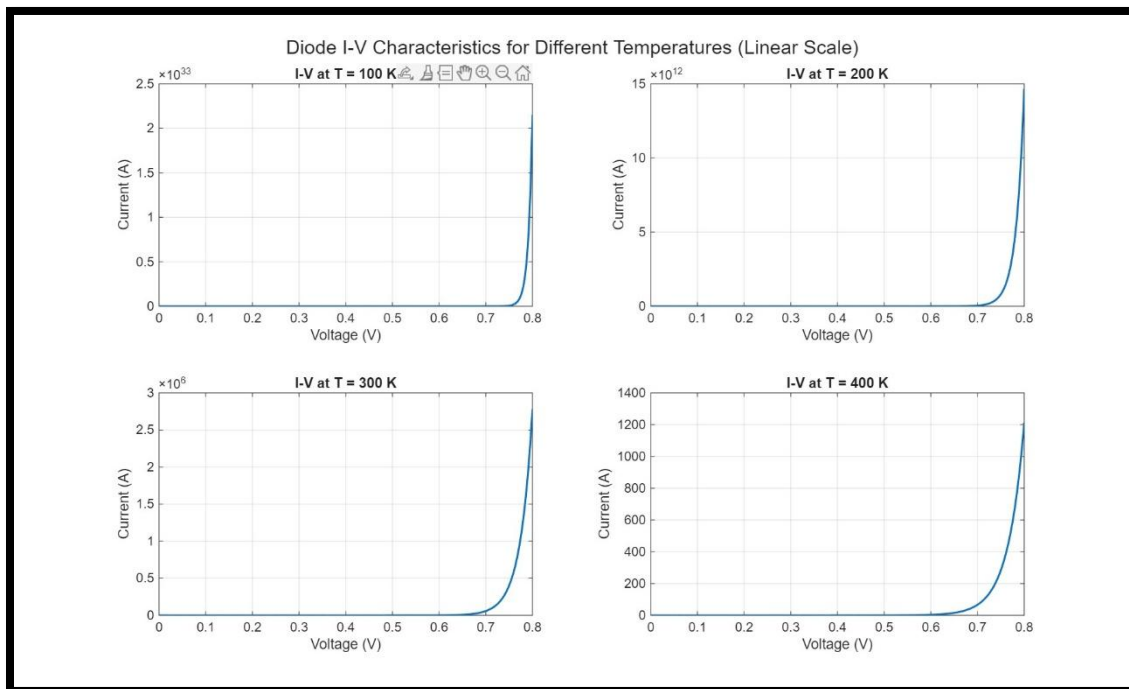


Figure 2: Ideal Diode Liner I-V Curve $T = 100K$ to $T = 400K$

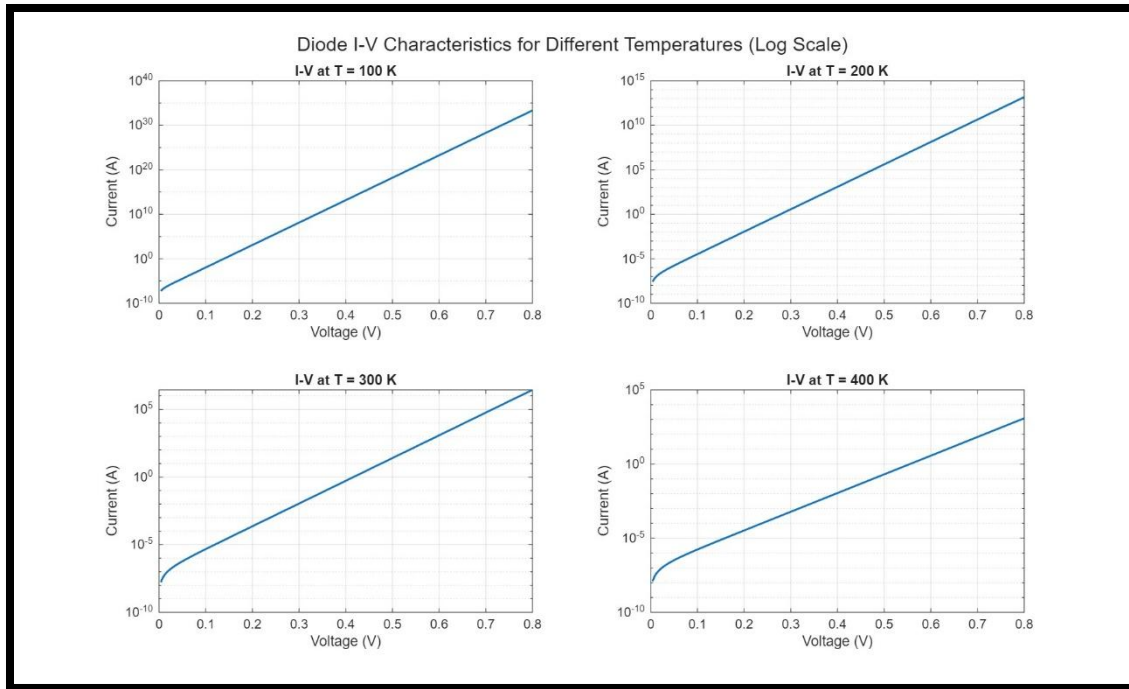


Figure 3: Ideal Diode Logarithmic I-V Curve $T = 100\text{K}$ to $T = 400\text{K}$

Plotting the non-ideal diode equation I-V curve for $T = 100\text{K}$ to $T = 400\text{K}$ and with $n = 1$ to $n = 5$ is next. I used a logarithmic y-axis, so it is easier to read.

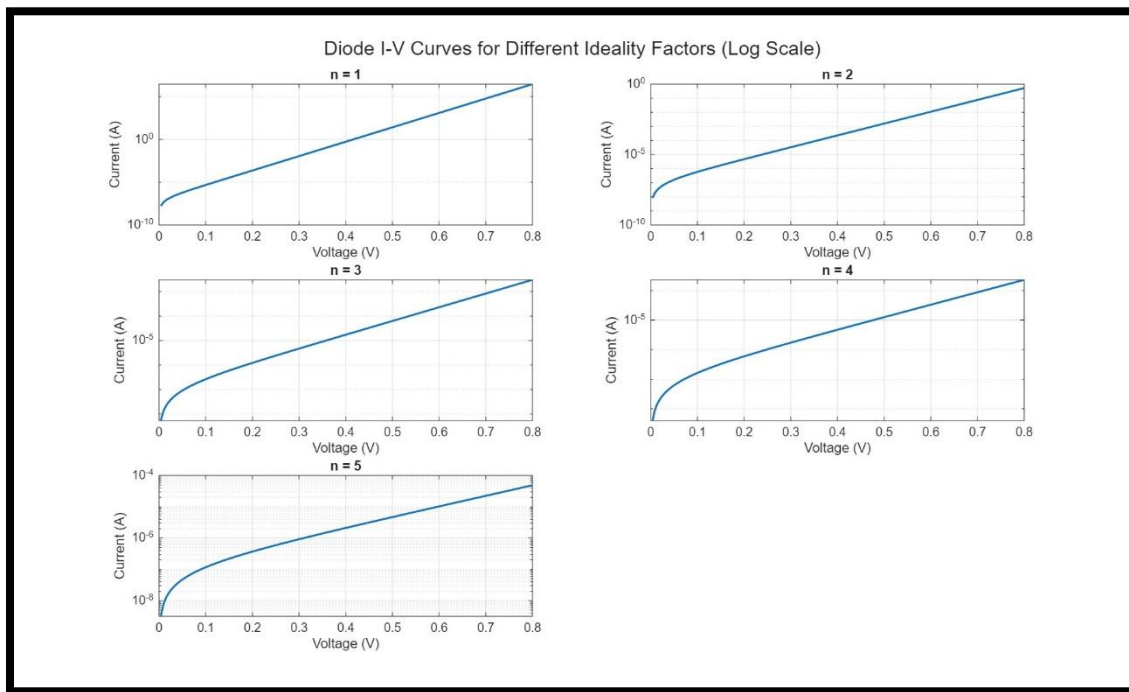


Figure 4: Non-Ideal Diode Logarithmic I-V Curve $T = 100\text{K}$ to $T = 400\text{K}$, $n = 1$ to $n = 5$

Experiment 1: Characterizing the I-V relationship of a small signal diode

The first experiment serves to understand the 1N4148 small signal diode. This is done by recording measured current and voltage data and plotting the results to be analyzed. The measured curve is compared to the ideal and non-ideal diode equations. This will explain the behavior of the diode. The 1N4148 small signal diode is connected to a power supply and digital multimeter. Throughout this lab, issues were encountered when utilizing a breadboard. So, the power supply and digital multimeter were connected directly to the diode to help eliminate inconsistencies. The I - V characteristics were measured in forward bias, and the current was measured in series from a multitude of voltages with the digital multimeter.

The diode's current was measured from 0 V to 1.1 V to observe the behavior. The data was plotted using MATLAB as shown below.

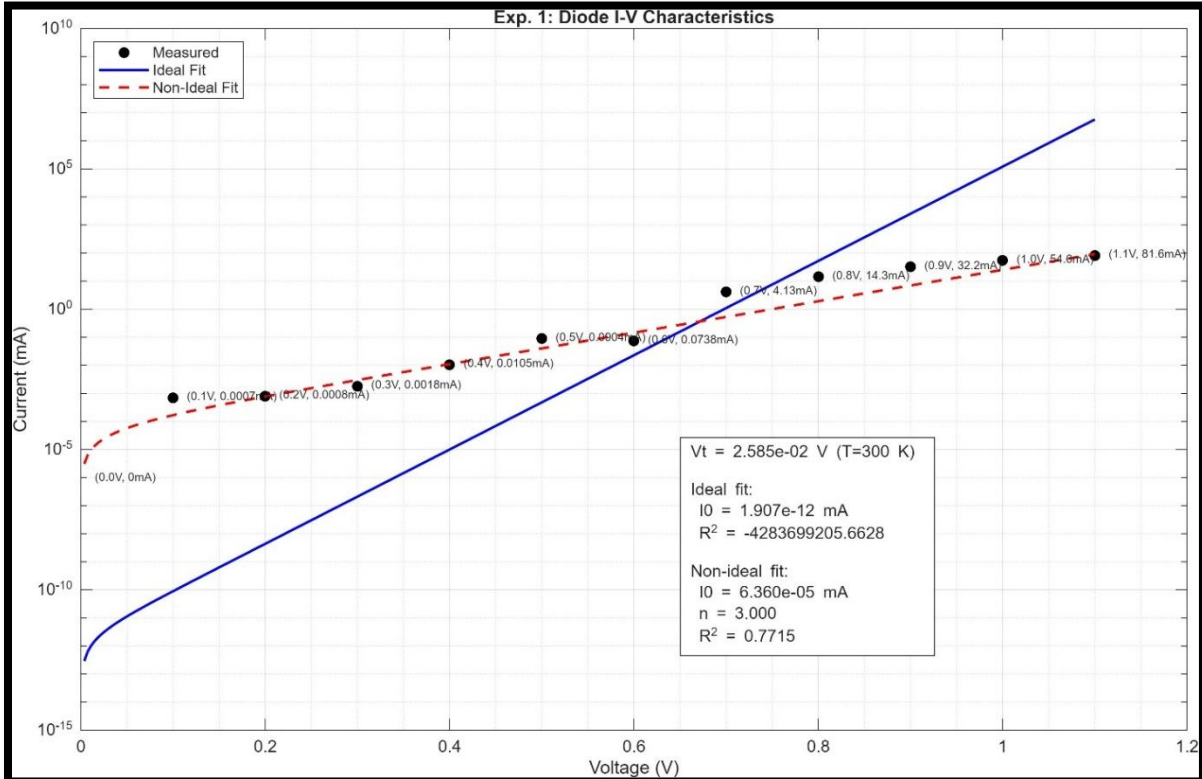


Figure 5: 1N4148 Diode I-V Characteristics

The black points plot the measured current corresponding to the voltage. The dotted red line represents the non-ideal diode equation, and the solid blue line represents the ideal diode equation. Values I_0 and n are shown in the text box on the graph. The measured current values behave somewhat as expected. The diode was only measured in forward bias, so $V > 0$. From 0.2 V to 0.7 V, the current rises, but it is extremely low. A more ideal diode should have a more exponential curve. In addition, there should be a substantial increase in current around 0.7 V, which occurs as well. Beyond that, the current should increase linearly as supported by the recorded measurements.

There are, however, a few issues with the measurements. At the lower voltages, 0V to 0.6V, the current is not perfectly exponential. This can be due to issues ranging from noise in the digital multimeter to a faulty diode. From the plot, it is clear to see that this diode follows the non-ideal diode fit more closely than the ideal. This is not necessarily a bad thing as it is uncommon for a diode to be mostly ideal. The calculated n value is unusually high as well. An ideal diode should have a value of n closer to 1.0. The higher value indicates a more non-ideal fit and less exponential behavior, which can be clearly observed. This can be caused due to various inconsistencies in our measurements but is most likely the less satisfactory behavior of this diode. Before settling on using this diode and its measurements, three others were tested, and each gave different current measurements to the previous. This only supports the fact that no diode will be perfectly ideal.

The value of I_0 for the non-ideal equation was calculated to be $6.360 \times 10^{-5} \text{ mA}$. If this is compared to the 1N4148's datasheet for its reverse current, I_R , then the calculated I_0 fits within specification. The Shockley Diode equation can be to approximate the diode's behavior as well.

Plugging in the non-ideal I_0 and n value, the results line up with the slight exponential behavior this diode exhibits.

Experiment 2: The resistor-diode series circuit

This second experiment introduces a resistor in series with the 1N4148 diode. The goal of this experiment is to see how a resistor changes the behavior of the diode and see how the circuit behaves at a range of voltage values. Like experiment one, a digital multimeter meter was connected to the resistor-diode circuit in series to measure current. A power supply was used with a current limitation of 200 mA. The first round of measurements did not come out as expected, so the breadboard being used was set aside and the circuit was wired directly to the digital multimeter and power supply. This provided much more consistent results. The measurements are plotted below utilizing MATLAB.

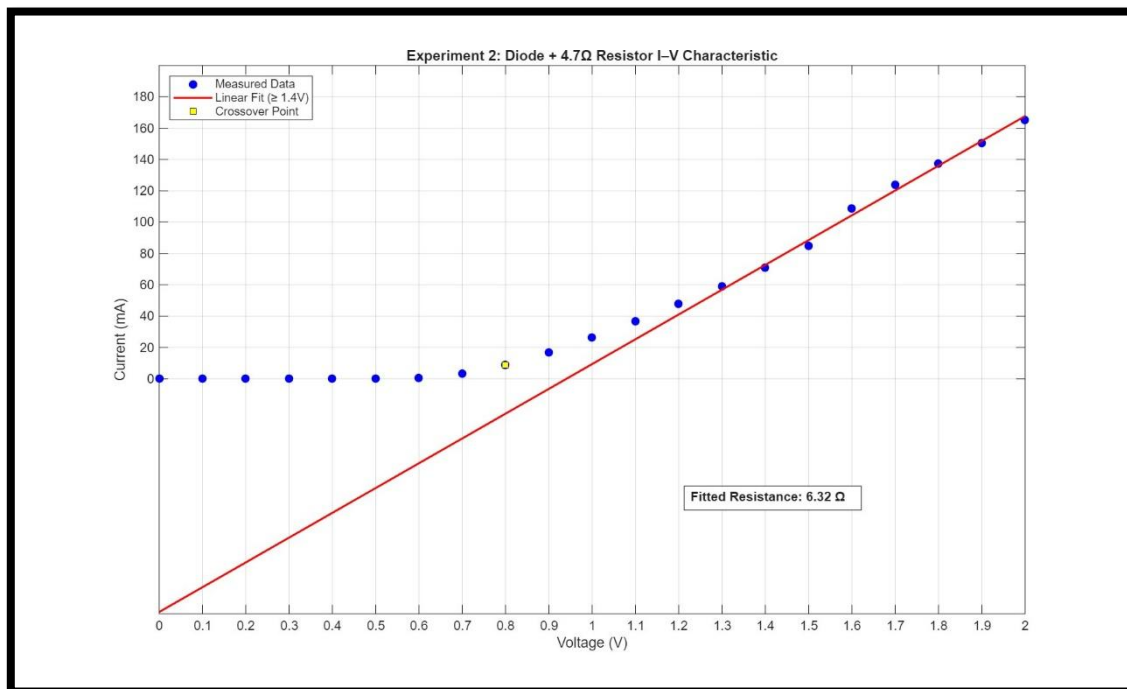


Figure 6: 1N4148 Diode with a 4.7Ω Resistor I-V Characteristics

The measurements show that at higher voltages, the diode starts to fully allow current to flow and the I-V curve becomes nearly linear. But at lower voltages, although very small, the current does increase exponentially until the crossover point shown in the graph. That crossover threshold is around 0.8 V, meaning the circuit starts to act in a more resistor-like behavior past that point. I am curious to see the differences in measurements if a much higher resistor was used with a more durable diode. The calculate resistance comes out to be 6.32Ω compared to the actual chosen value of 4.7Ω . This is a remarkably close value considering the resistor used has a 10% tolerance.

A rigorous way to determine the crossover point can be done by utilizing these equations with clever substitutions:

$$\text{Dynamic Resistance of the Diode: } r_d = \frac{nV_T}{I}$$

$$V_T = \frac{kT}{q}$$

Defining the crossover current to be where the dynamic resistance meets the series resistor: $r_d(I_C) = R$

$$\text{Solve for the Crossover Current: } I_C = \frac{nV_T}{R}$$

I_C is where the diode and resistor each have an effect on the circuit at the same time. Defining it as the crossover current. To find the corresponding voltage, I_C can be substituted into Shockley's diode equation and then solving for V_d .

Experiment 3: Characterizing a Zener Diode

The third experiment serves to understand the reverse-bias behavior of the 1N5226 Zener diode. The breakdown voltage, V_Z , and dynamic resistance, R_Z , is calculated. Plotting the I-V curve using the recorded measurements from forward and reverse bias, the behavior of the Zener diode can be understood.

A 1N5226 Zener diode was connected in series to a digital multimeter, same as experiment one. Measurements were recorded with voltages from a power supply ranging from -5 V to +5 V in 0.5 V increments. No resistor or breadboard was used between the connections, as inconsistent results were found when trying to do so. The power supply's current limit was set to 200 mA to prevent potential burning of the diode.

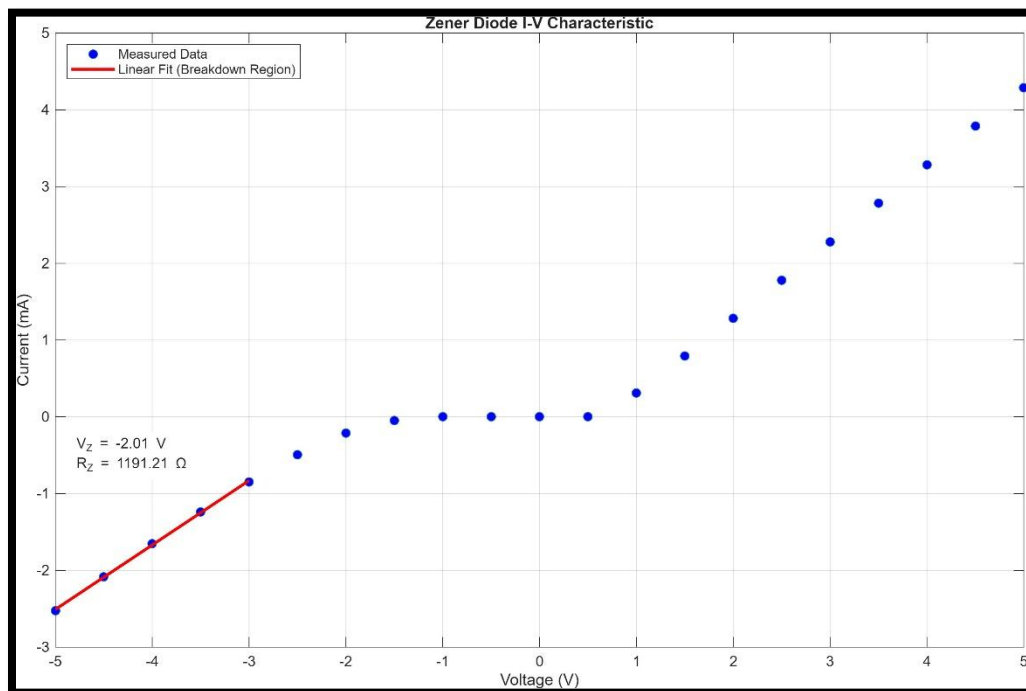


Figure 7: 1N5226 Zener Diode I-V Characteristics

The breakdown region occurred from $V < 3.5$ V. This is observed in the plot as the current increased relatively linearly with each new step in voltage. Plotting a straight line to these values yielded a V_Z value of -2.01 V. In forward bias, the diode's current increased exponentially,

as expected, starting at ~ 0.7 V. These results line up with Shockley's diode equation. There should not be a substantial change in current until breakdown, then remain about a steady rate until forward bias when it will exponentially increase. These measurements represent a typical diode I-V curve. The small discrepancies with the change in current from around -2 V to 0.5 V cannot be seen in the graph. But the values shift up and down a few mA which can be attributed to minor deviations in the diode's temperature and time between measurements. However, this data is largely sound with what is to be expected.

The circuit from part a) of this experiment works because Zener diodes should maintain a constant voltage when in its reverse breakdown state. Measuring the voltage while applying the reverse bias will show whether it is a Zener diode and its V_Z . This is because a regular diode will output the full reverse bias voltage. This diode's V_Z experimental measurement came out to -2.01 V. The 1N5226's datasheet shows that it should be around -3 V to -3.3 V. Although the value is low, the diode behaves as expected and no major discrepancies are seen. I believe this 1 V difference in the expected value can be fixed by using a resistor in series, limiting the power supply source current further, and plotting in smaller increments to gather more data. The I-V curve from -5 V to -3 V is very linear. It provides an R_Z of 1191.21Ω , the resistance of the diode when in its breakdown state. This is quite a high value for R_Z meaning the diode voltage changes significantly more with a higher load than a lower R_Z value. I believe this can be corrected if measured with higher current and a resistor in place. However, measurements are consistent within themselves, so this value appears accurate.

Experiment 4: The half-wave rectifier

Experiment four serves to illustrate how a half-wave rectifier works as it takes an AC signal, such as a sine wave, and converts it into a DC signal. Doing so allows the observation of a diode only allowing current to flow in one direction. This is done with the two previous diodes used in this lab, the 1N4148 and the 1N5226 Zener diode.

The setup involves a diode placed in series with a resistor, in this case a $4.7\text{k}\Omega$ load resistor, an AC voltage source moving across the diode and resistor, and a voltage measurement over the resistor. A 1kHz to 5kHz sine wave was sent through the circuit and an oscilloscope was used to take measurements. A 1kHz wave was used to start and was increased up to 5kHz in increments of 1kHz. The input wave is shown below:

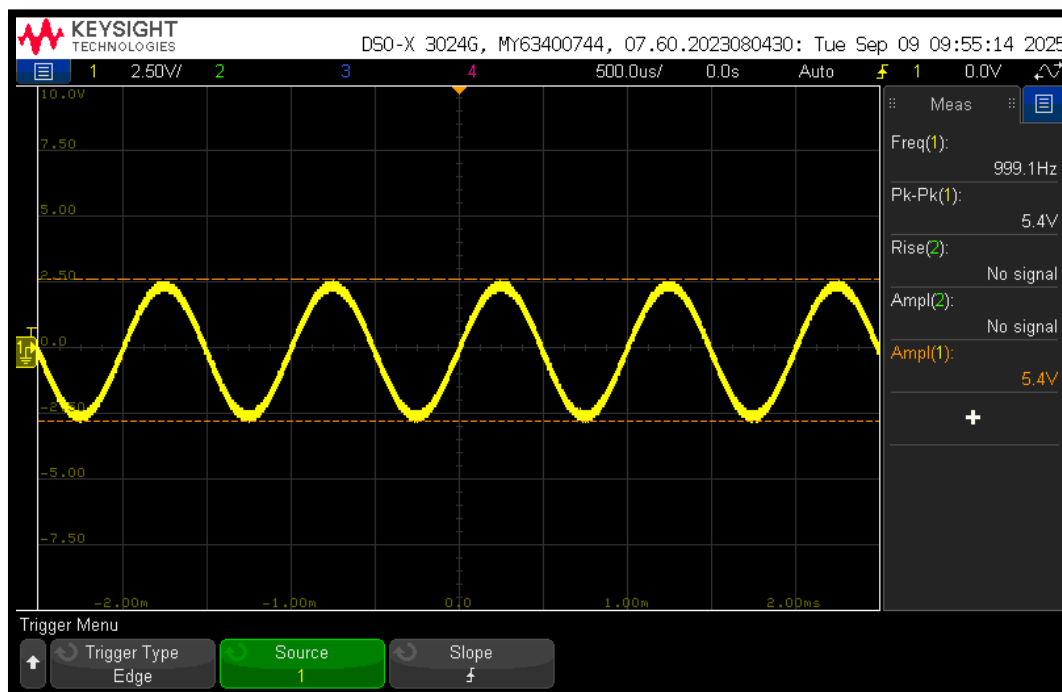


Figure 8: 1kHz AC Sine Wave Input

1N4148 Diode Output Measurements

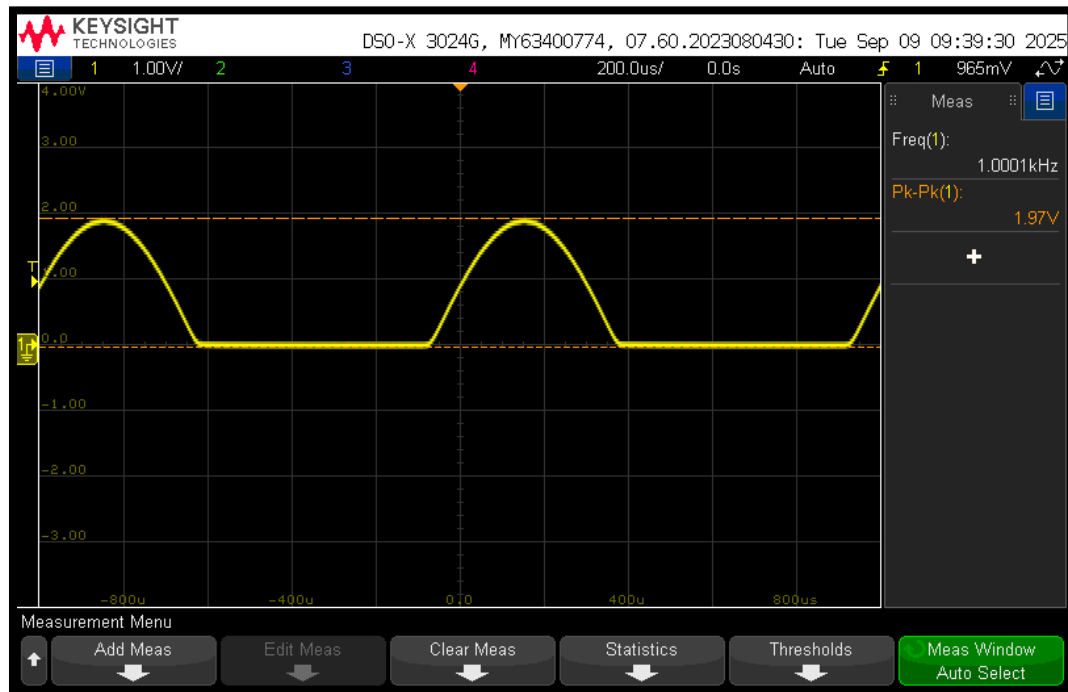


Figure 9: 1kHz DC 1N4148 Output

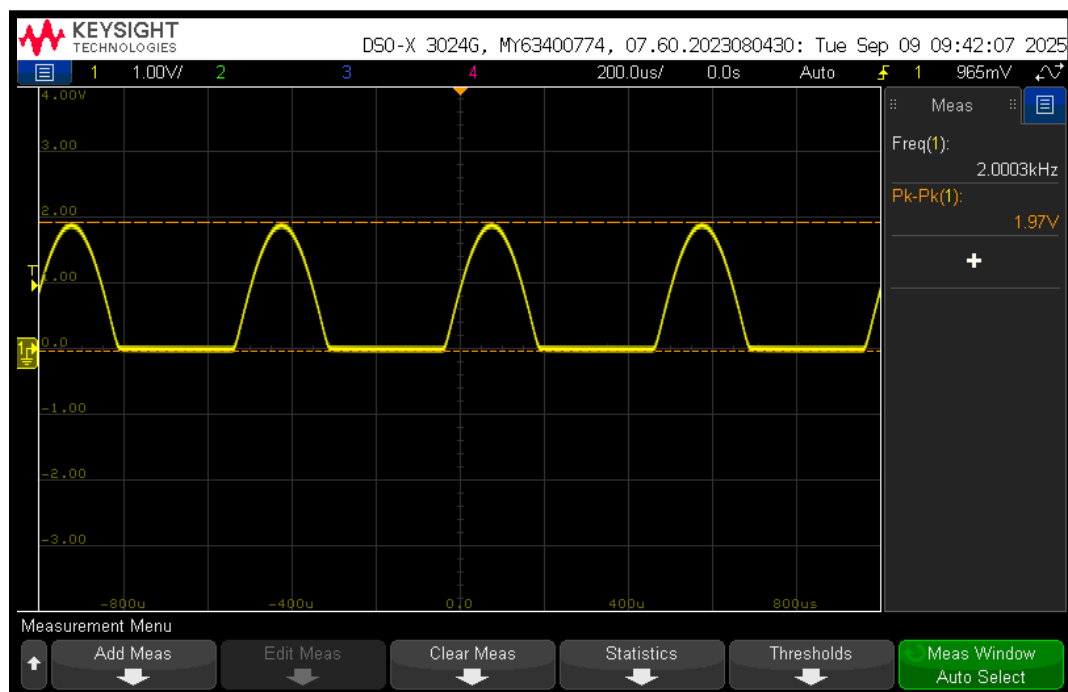


Figure 10: 2kHz DC 1N4148 Output

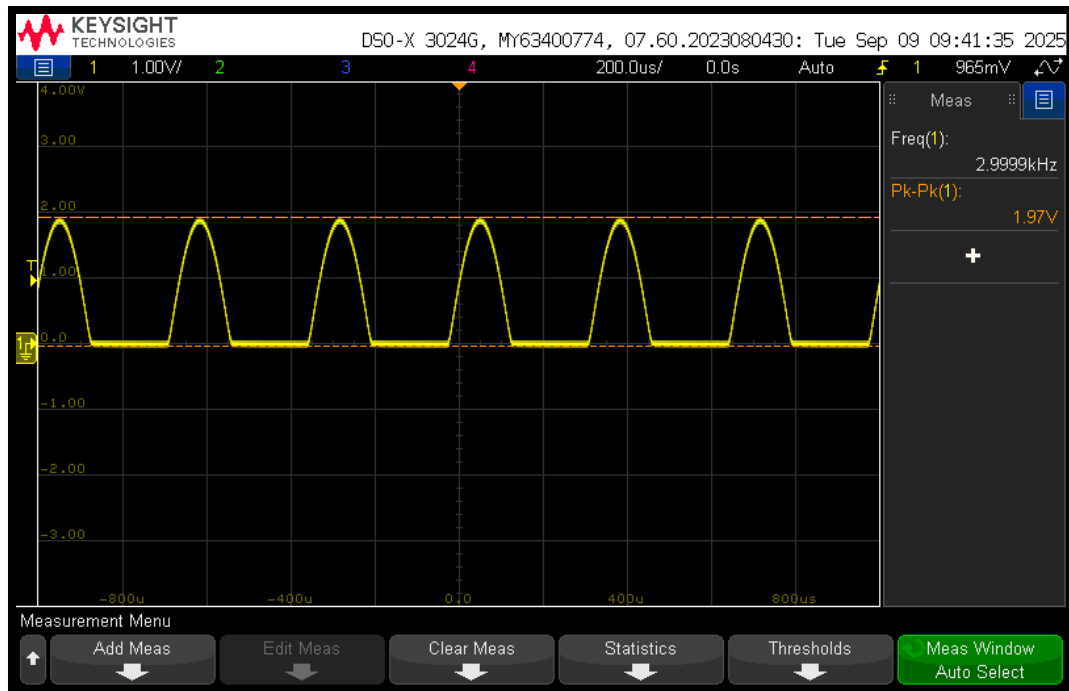


Figure 11: 3kHz DC IN4148 Output

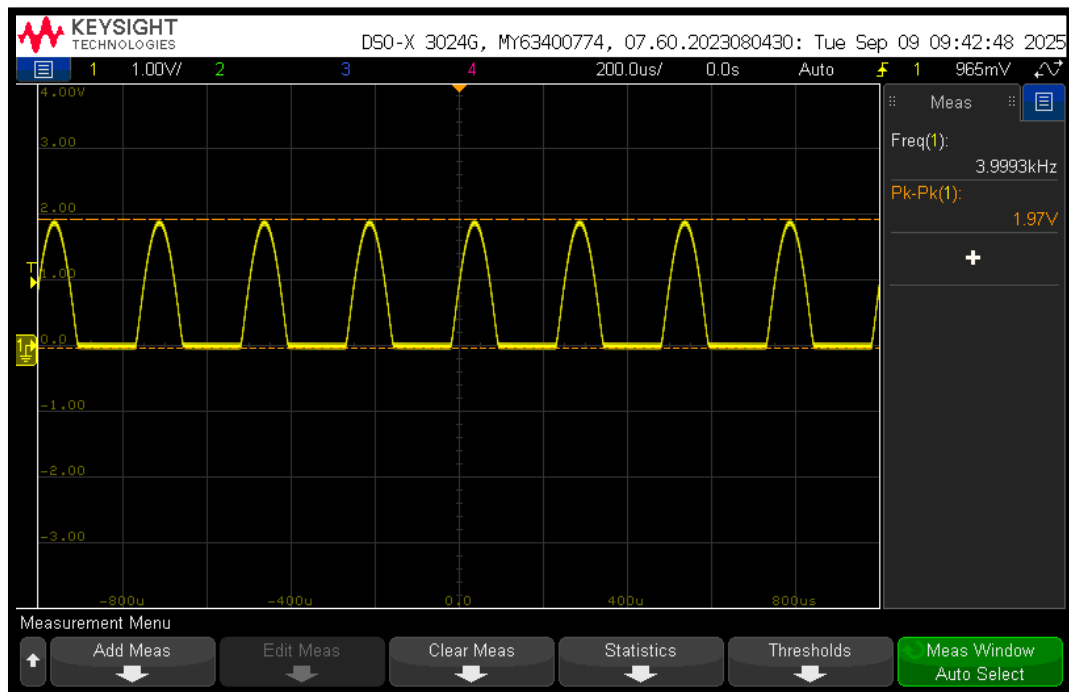


Figure 12: 4kHz DC IN4148 Output

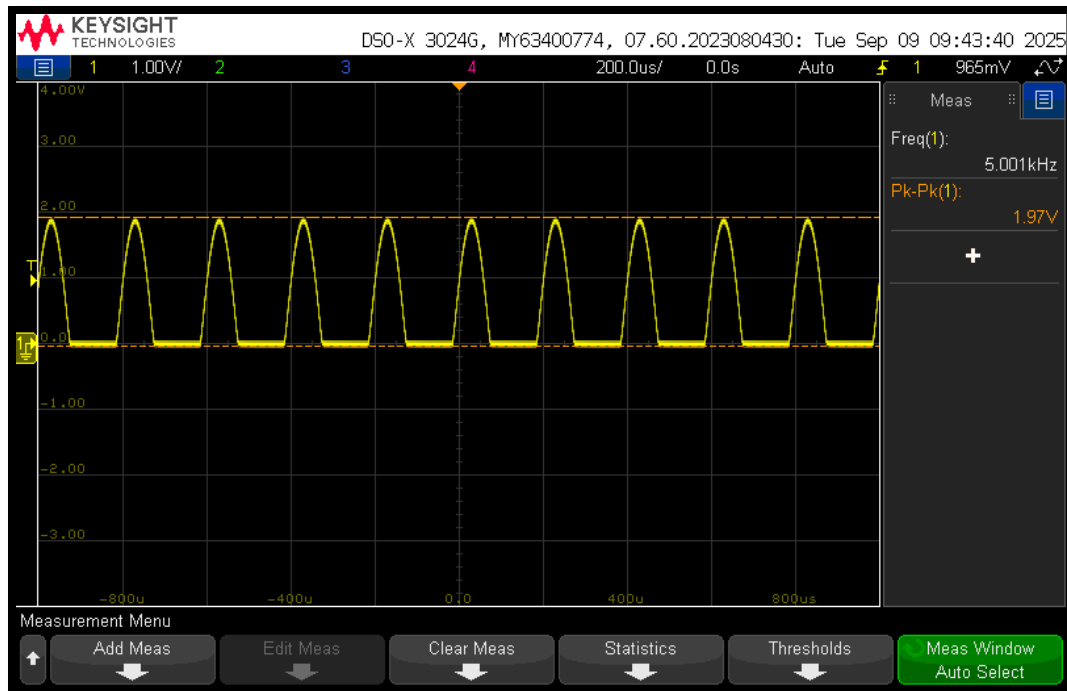


Figure 13: 5kHz DC 1N4148 Output

The frequencies going into the circuit from the AC sine wave remain the same in the DC output. However, due to being a DC signal, there is no reverse voltage. The DC output should look like the positive side of its AC wave counterpart. This is especially evident by comparing then 1kHz AC wave with the 1kHz DC wave. The frequency remains the same with only a small voltage drop on the DC output. The 1N4148 diode switches quickly which is why there is little distorting of the DC wave.

1N5226 Diode Output Measurements

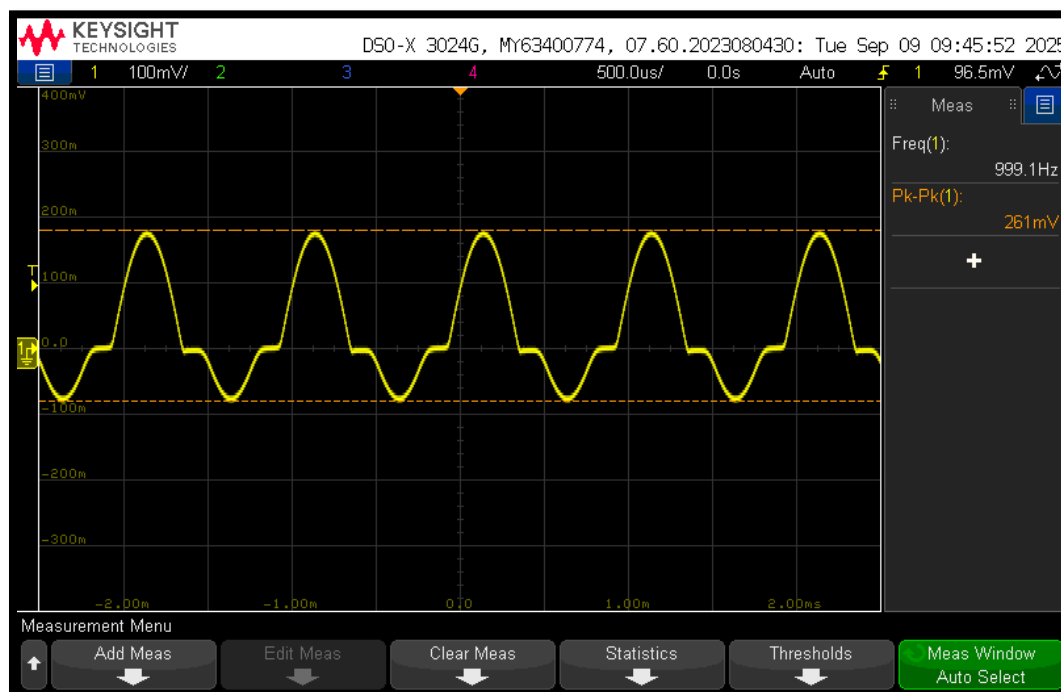


Figure 14: 1kHz DC 1N5226 Output

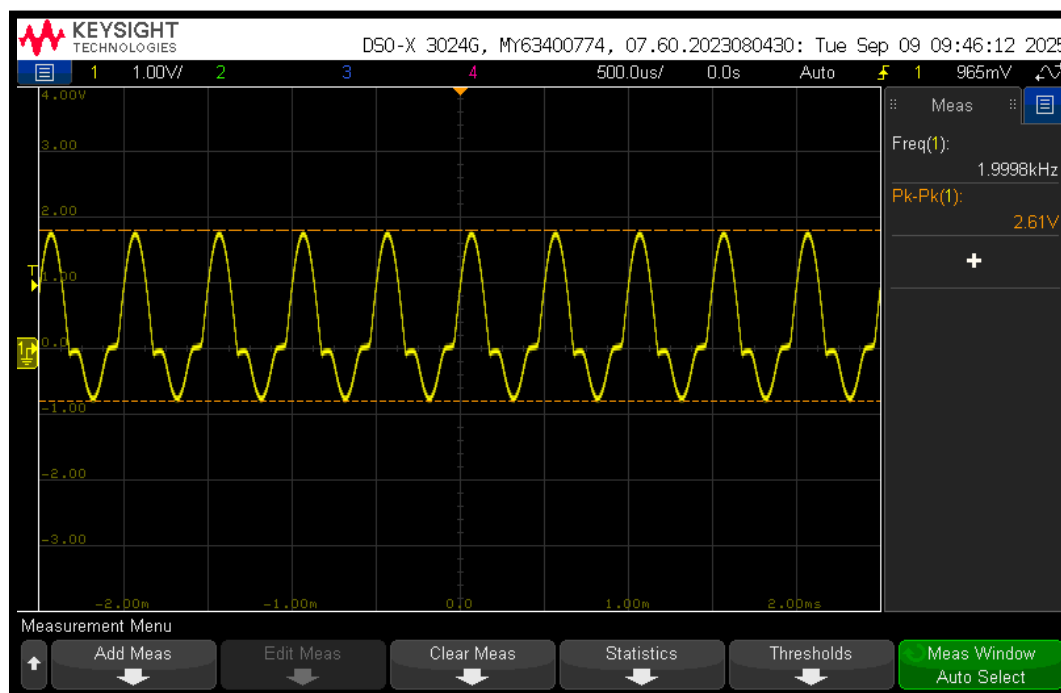


Figure 15: 2kHz DC 1N5226 Output

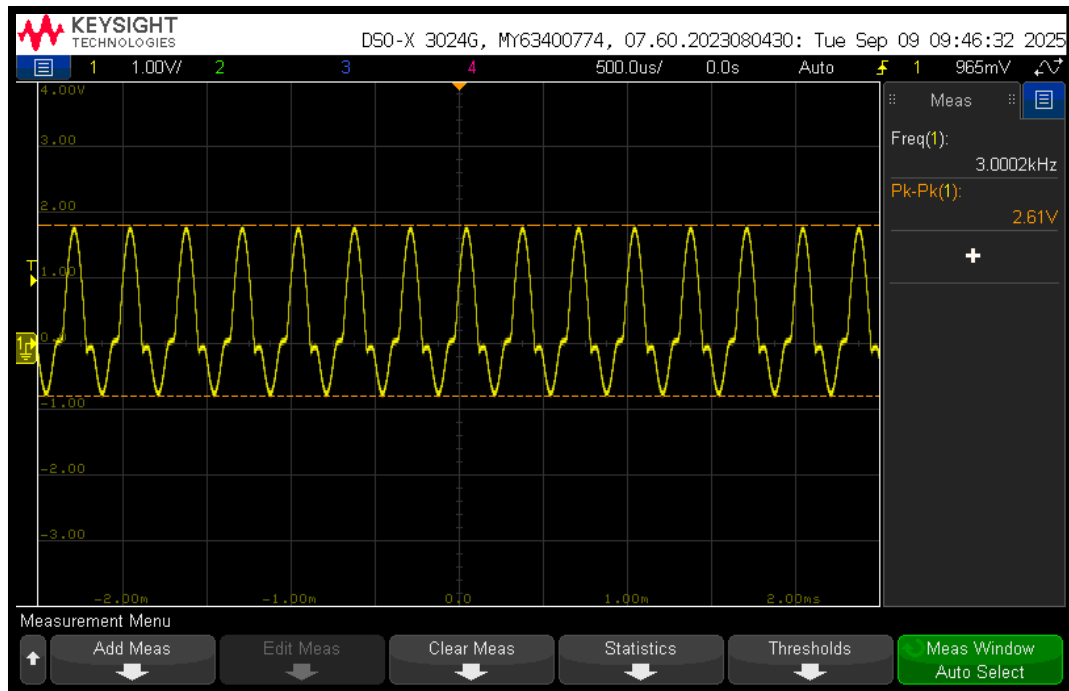


Figure 16: 3kHz DC 1N5226 Output

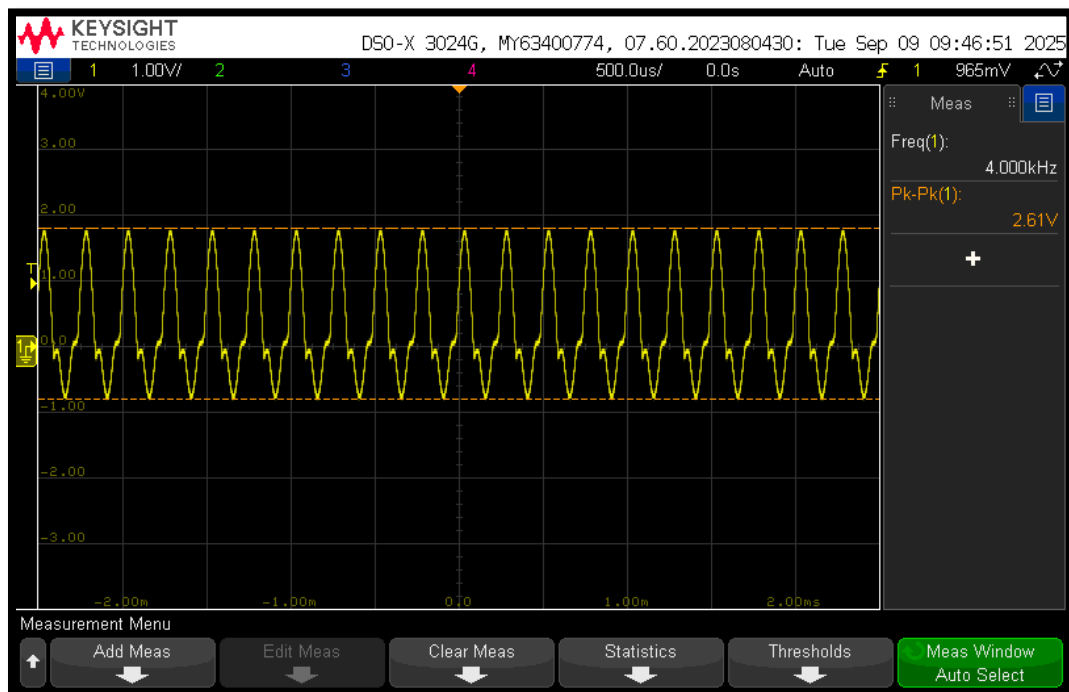


Figure 17: 4kHz DC 1N5226 Output

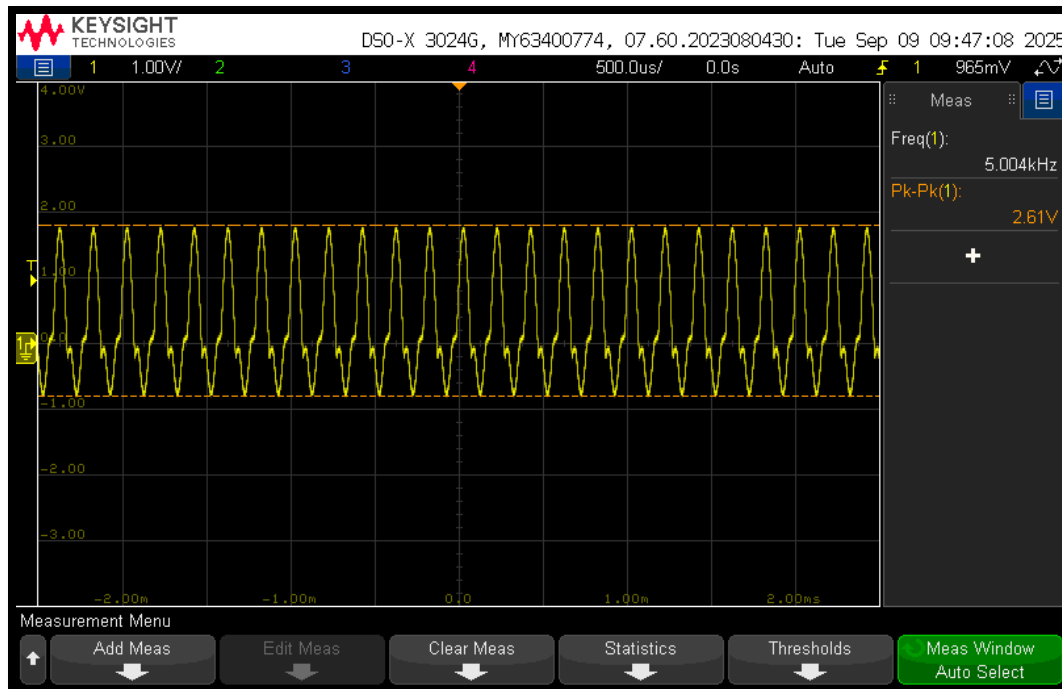


Figure 18: 5kHz DC 1N5226 Output

The frequencies going into the circuit from the AC sine wave look different compared to the DC output. The 1N5226 is a much slower diode compared to the 1N4148. This is evident at much higher frequencies, it starts to be more noticeable at anything above 3kHz. The diode cannot switch as fast adding more distortion in the wave. This is most noticeable at 5kHz and there is still a slight negative voltage reading on the oscilloscope in every measurement.

A half-wave rectifier should only allow current through the positive side of the cycle resulting in an on and off DC output. With these measurements this holds true, but with a little inaccuracy on the Zener diode. However, if started at lower frequencies, both diodes should produce comparable results. The diode should only conduct with a positive voltage blocking current from moving in the opposite direction (forward bias). This process converts the AC input to DC output. There are many applications for this. Power supplies and electricity transportation utilize this method. The standard US house will deliver AC current out of its outlets. Lower

powered household devices filter the current into DC to be utilized. The main difference between the behaviors of the two diodes is due to their switching times. The 1N4148 diode is much quicker and manages the 5kHz wave far easier compared to the slower 1N5226 diode. The 1N5226 has a higher capacitance and therefore longer recovery time.

Simulation

A PN junction simulation was run with different parameters to observe the behavior of diodes. I kept the material as Silicon as that is the material of the two diodes used throughout this lab. I ran one simulation for default parameters, one for lower acceptor concentration with higher donor concentration, one for higher temperature, and one for lower temperature. The output I-V curves are below:

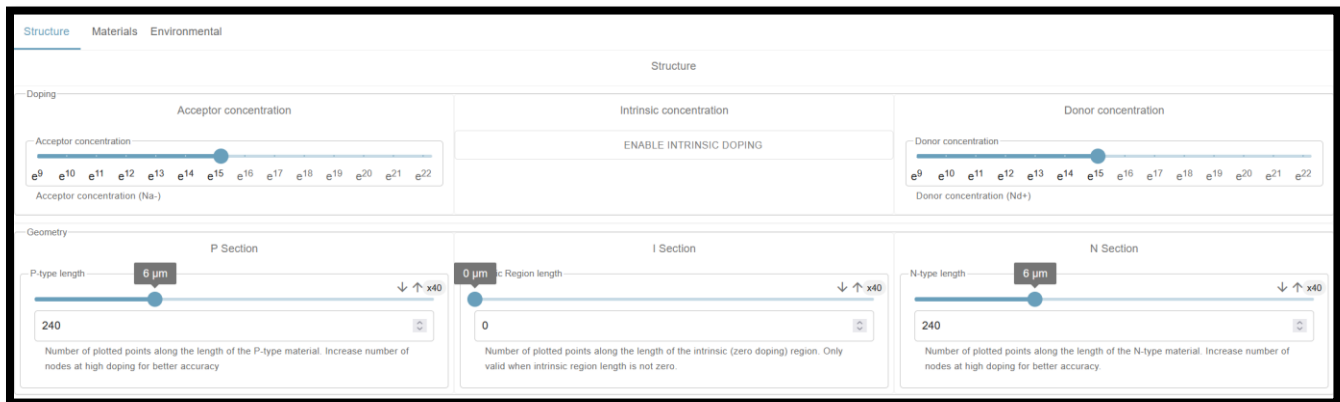


Figure 19: Default Parameters with Silicon Material and $T = 300\text{K}$

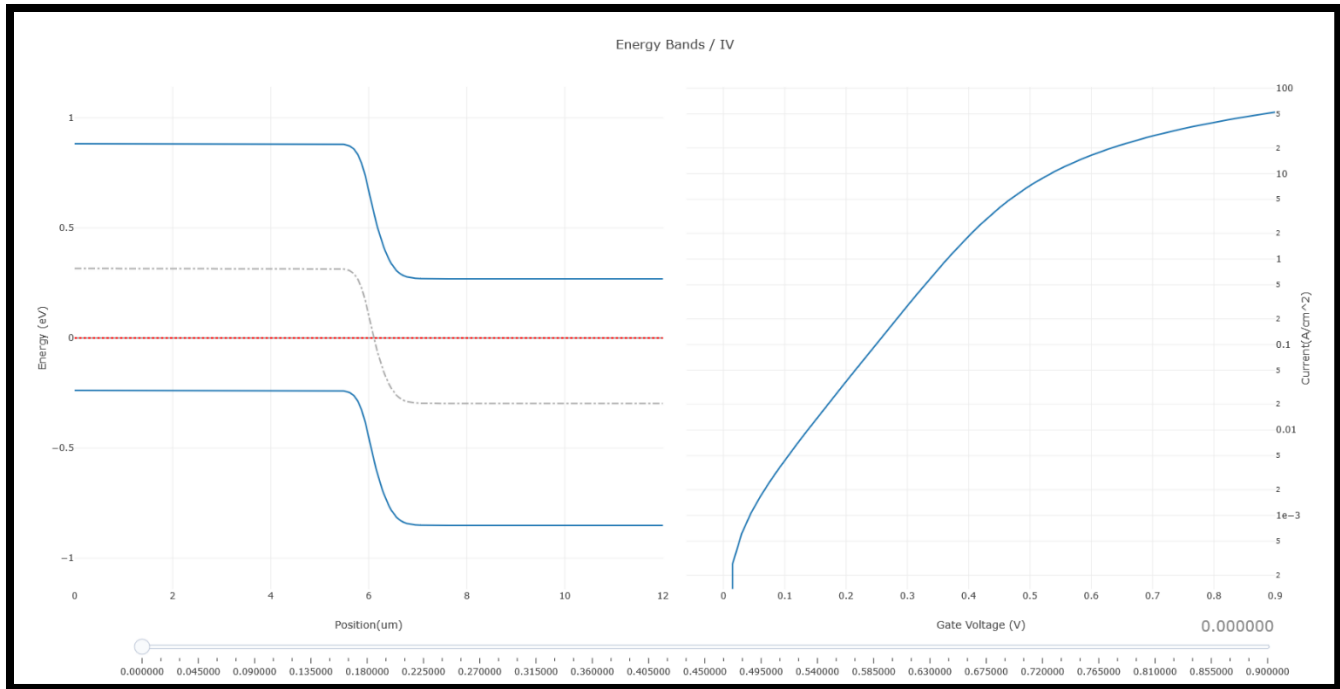


Figure 20: I-V Curve for Default Parameters with Silicon Material and $T = 300\text{K}$

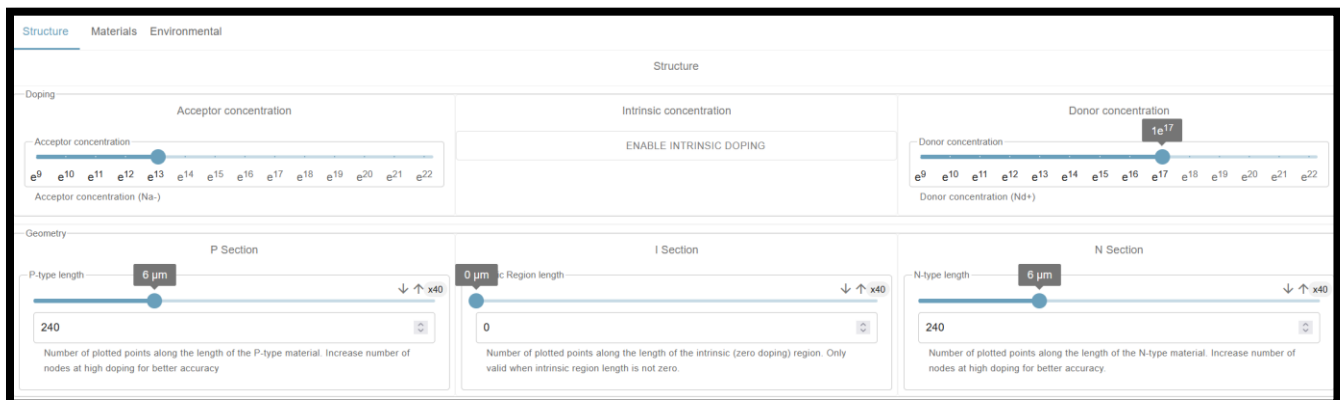
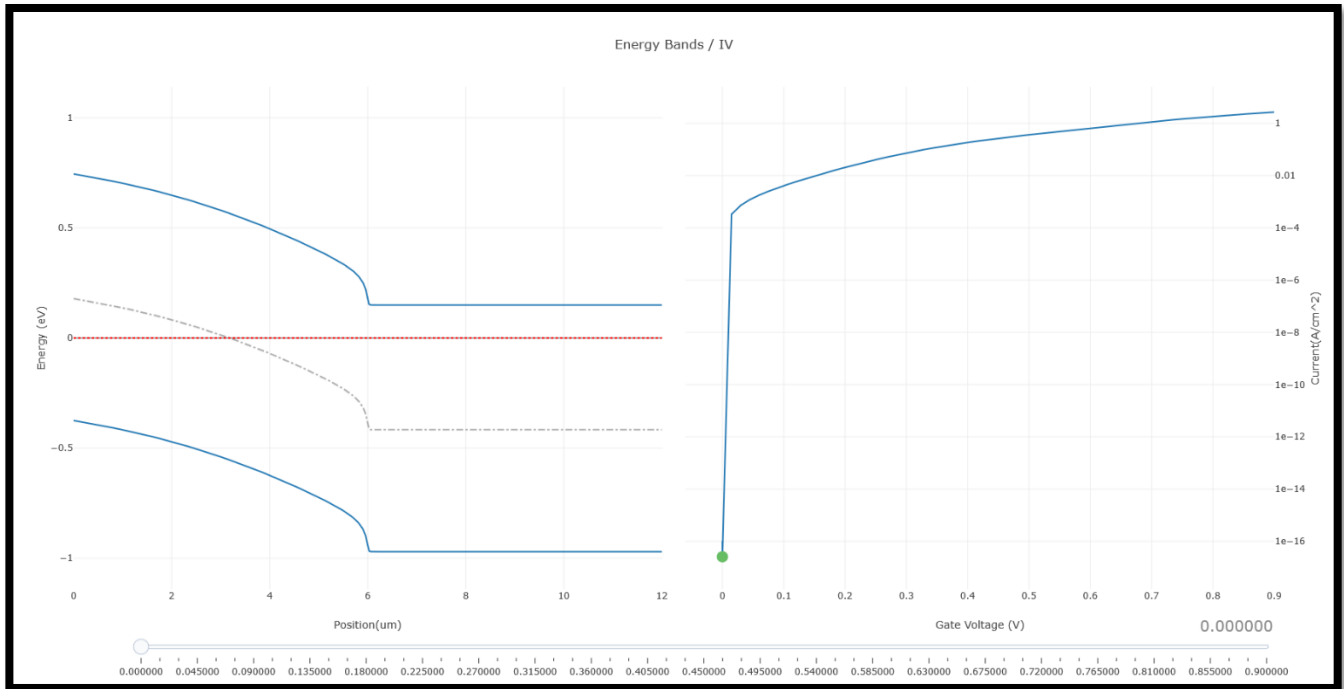


Figure 21: Less Acceptor Concentration and More Donor Concentration with Silicon

Material and $T = 300$



*Figure 22: Less Acceptor Concentration and More Donor Concentration with Silicon Material
and $T = 300K$*

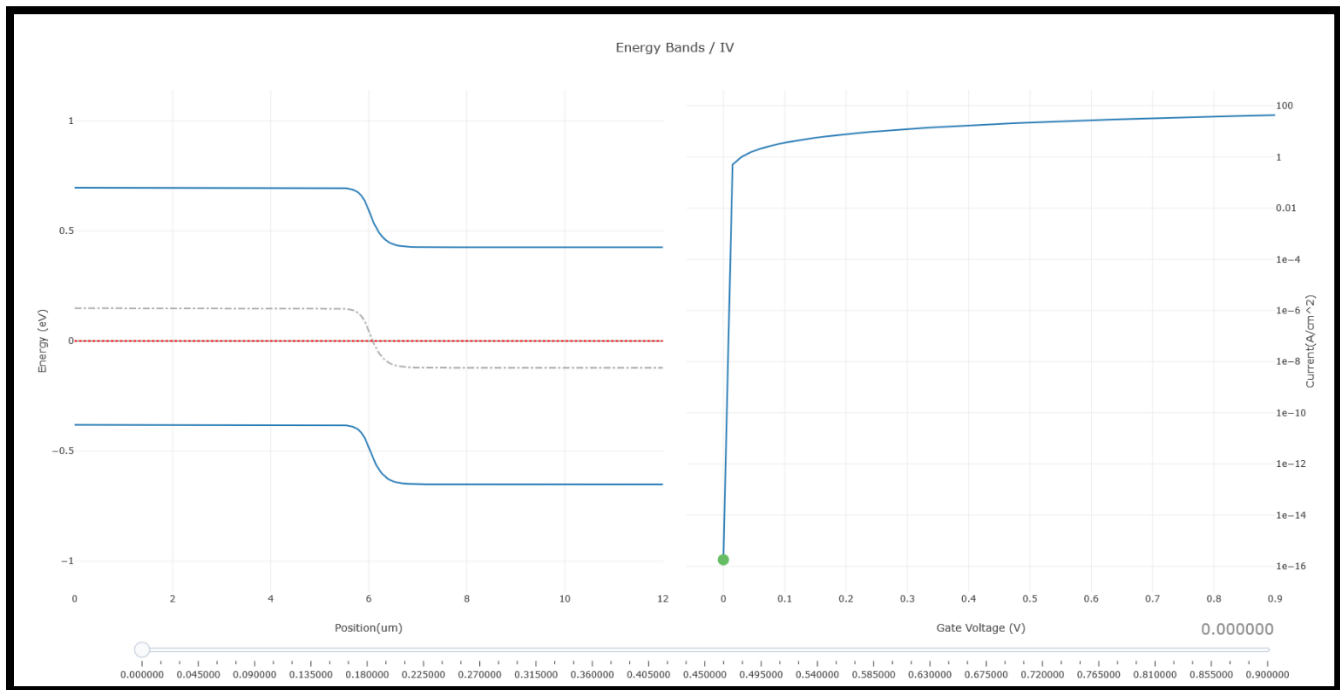


Figure 23: Default Parameters with Silicon Material and $T = 450K$

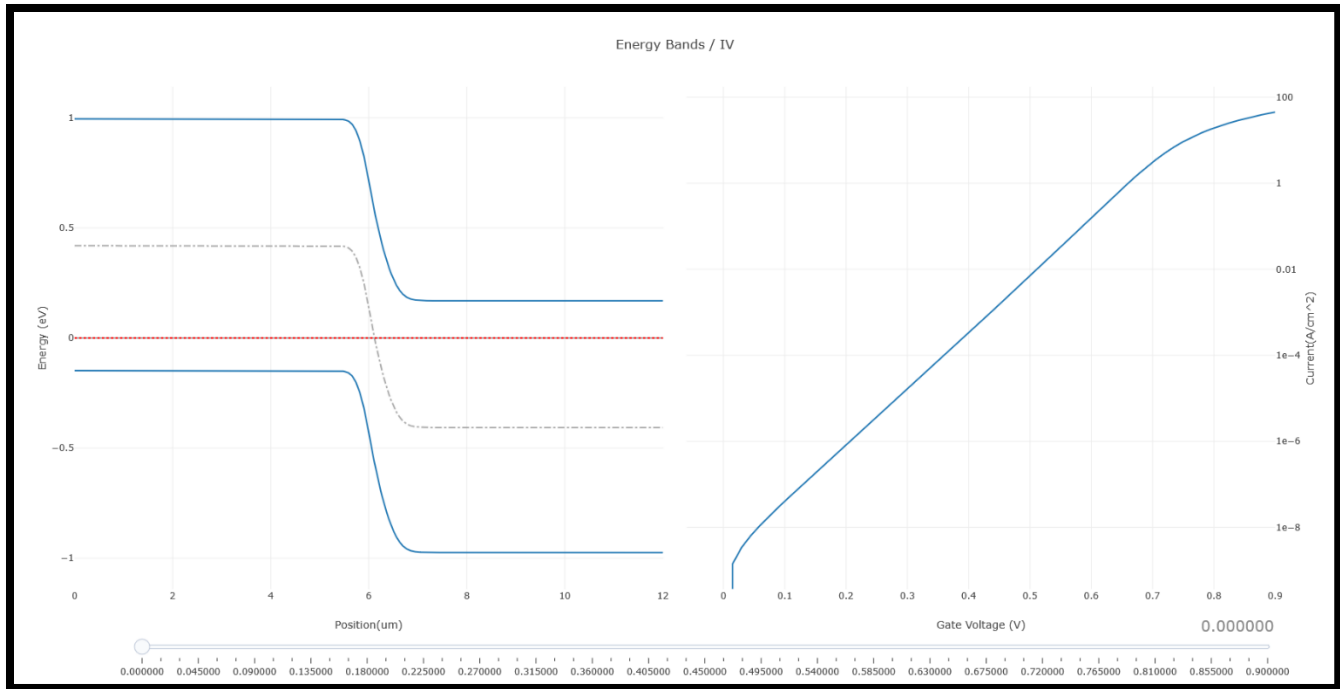


Figure 24: Default Parameters with Silicon Material and $T = 200K$

For the default parameters, the behavior is perfectly as expected. The second simulation was run with a decreased acceptor concentration and an increased donor concentration. I was curious to see what the results would come out to be. I expected the results to be uneven with such an imbalance of concentration. The I-V curve should still be exponential, but the lower acceptor concentration will make this far more abrupt. The graph produced is odd looking but does fit what should be expected. Lowering the acceptor concentration while increasing the donor concentration appears to move the depletion region to the heavier doped side. However, I was most interested in how temperature would change these results. I ran a simulation for default parameters for a Silicon diode but with $T = 450K$ and $T = 200K$. I would expect a lower temperature to have a far lower current than the higher temperature. I would also expect a higher temperature to change the threshold of the diode. Looking at the results, this happens as well. The higher temperature diode activates far earlier than the lower temperature diode. The $T =$

450K diode also produces a larger current value at a much lower voltage compared to the T = 200K diode.

Conclusion

After observing the behavior of diodes in these four experiments, it becomes clear that they serve a significant role in the modern electrical world. Their ability to control current allows for voltage regulation, switching, circuit protection, and current conversion. A standard and Zener diode's I-V curves were compared to their theoretical behaviors. The breakdown voltage V_Z was measured for the 1N5226 Zener diode proving the existence of a breakdown threshold. The data supports a Zener diode's ability to regulate voltage once that breakdown threshold has been achieved. More importantly, my MATLAB skills have improved, and I have learned to better analyze data and go back and correct if something does not measure as expected.

Citations:

Introduction diode structure image:

Fluke Corporation. "What Is a Diode?" *Fluke*, 2025, <https://www.fluke.com/en-us/learn/blog/electrical/what-is-a-diode>